

Analyzing Traffic Stop Citation Patterns Across Chicago Neighborhoods

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1. Introduction

1.1. Project Background

The Chicago Police Department (CPD) has faced longstanding scrutiny regarding disparities in traffic stop practices, with prior research documenting differences in how stops are conducted across communities. In addition to individual-level factors, recent work highlights the role of neighborhood-level socioeconomic conditions in shaping policing outcomes. For example, Kirk (2008) identifies concentrated poverty and collective efficacy as important predictors of crime and enforcement patterns, noting that communities with higher poverty levels often experience different policing dynamics.

This motivates our use of the Chicago Health Atlas dataset, which provides community-level socioeconomic indicators, allowing us to examine how broader neighborhood characteristics relate to variation in traffic stop outcomes rather than focusing on individual-level attributes.

Recent studies, such as Xu et al. (2024), have combined policing and geographic data to analyze patterns in traffic stops, demonstrating that contextual factors like population density and neighborhood composition play an important role. Building on this work, we adopt a similar data integration approach but focus specifically on socioeconomic conditions at the neighborhood level to better understand how these factors are associated with variation in citation outcomes across Chicago.

1.2 Ethical Considerations

This project uses publicly available, deidentified data from the Stanford Open Policing Project (SOPP) and the Chicago Health Atlas. Our analysis is conducted at the community-area level, meaning all results are aggregated rather than tied to individual traffic stops or specific people. This reduces privacy concerns and ensures that no individual can be identified.

Because this project examines relationships between socioeconomic factors and traffic stop outcomes, careful interpretation is essential. We present our findings as associations rather than causal relationships and avoid making claims about individual behavior. The goal is not to assign blame, but to better understand broader neighborhood-level patterns and contribute to informed

discussion.

1.3 Project Objectives

This project focuses on understanding how citation likelihood varies across neighborhoods in Chicago using community-level socioeconomic data.

First, our descriptive objective is to examine how citation rates vary across community areas and to identify patterns in these rates across different socioeconomic conditions. We use summary statistics and visualizations to highlight meaningful differences across neighborhoods.

Second, our inferential objective is to assess how neighborhood-level socioeconomic factors are associated with citation likelihood. We use a grouped binomial logistic regression model to estimate these relationships and interpret how factors such as income, perceived safety, and trust in law enforcement relate to variation in citation rates.

Together, these objectives aim to move beyond simple description and toward understanding how broader neighborhood characteristics are associated with differences in traffic stop outcomes.

2. Data and Data Exploration

2.1. Data Description

Our project uses two publicly available datasets: the Stanford Open Policing Project (SOPP) Chicago dataset and the Chicago Health Atlas.

The SOPP Chicago dataset contains approximately 210,000 observations of vehicle and pedestrian stops conducted by law enforcement between 2011 and 2020. The dataset includes information on stop location (latitude and longitude), stop reason, date and time, and whether a citation was issued. Each row represents an individual traffic stop.

The Chicago Health Atlas provides community-level indicators across Chicago's 77 community areas. These variables capture socioeconomic and demographic characteristics, including measures of perceived neighborhood safety, trust in law enforcement, economic diversity, median household

income, and limited English proficiency. Each row corresponds to a community area rather than an individual observation.

2.2. Data Wrangling

To prepare the data for analysis, we first linked traffic stops to their corresponding community areas. Because the SOPP dataset only provides latitude and longitude coordinates, each stop was converted into a spatial point and joined to the Chicago community area boundary polygons using GeoPandas. Observations that did not fall within a defined community area were removed to ensure consistency with neighborhood-level variables.

Next, the Chicago Health Atlas data was merged with the traffic stop data using the community area identifier. All socioeconomic variables were converted to numeric format, and rows with invalid or missing values were removed to ensure compatibility with regression analysis.

We then aggregated the traffic stop data to the community level by computing citation rates within each community area. This step aligned the unit of analysis with the Health Atlas data, resulting in a dataset where each observation represents a single community area.

Finally, variables were selected based on interpretability and correlation analysis. Highly correlated predictors were removed to minimize potential multicollinearity, resulting in a final set of socioeconomic variables used in the regression model.

A key limitation of the original dataset is the vast amount of missing data in driver-level demographic variables, particularly race and age, among observations where a citation was issued. This imbalance creates the risk of biased or misleading results if these variables are included directly in individual-level models. To address this issue, we shifted our analysis to the community level, where socioeconomic indicators are more consistently measured. This approach allows us to examine broader patterns in citation outcomes while avoiding biases introduced by incomplete individual-level data.

2.3. Exploratory Data Analysis (EDA)

To support our descriptive objective, we explore how citation rates vary across community areas and whether these differences are related to socioeconomic conditions.

Figure A illustrates the geographic distribution of traffic stops across Chicago.

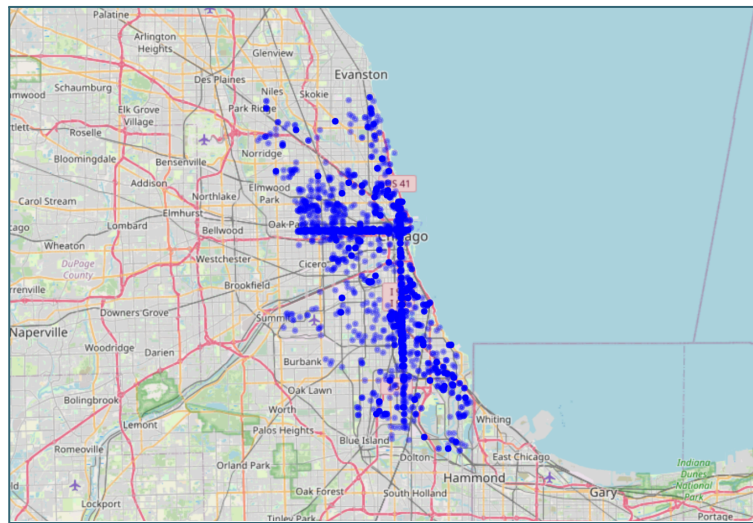


Figure A. Geographic distribution of traffic stops across Chicago. Each point represents an individual stop, with higher concentrations visible along major roadways and densely populated areas.

As shown in Figure A, stops are concentrated along major roads and densely populated areas.

However, this spatial distribution alone does not explain differences in citation outcomes.

To better understand variation in outcomes, Figure B shows the distribution of citation rates across community areas.

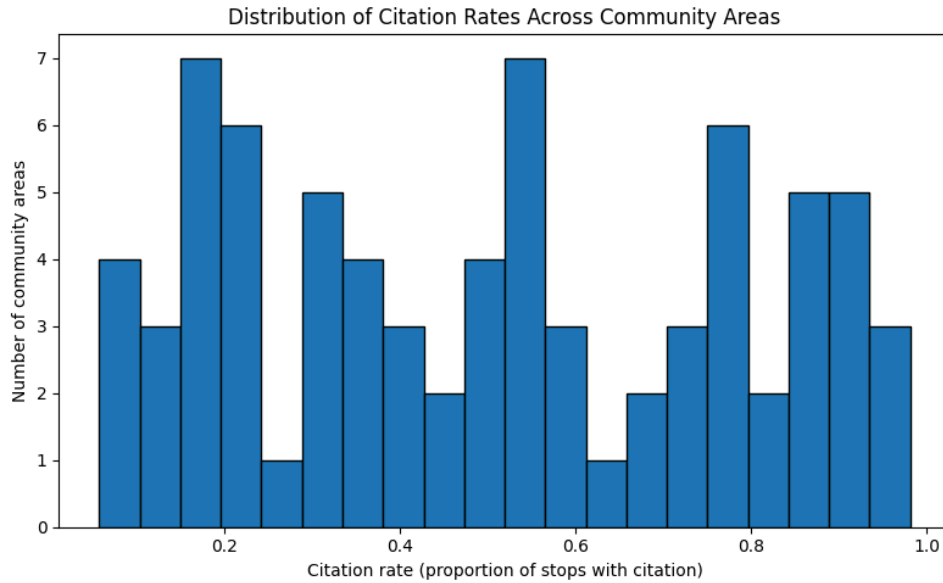


Figure B. Distribution of citation rates across Chicago community areas. Citation rates vary widely, indicating substantial differences in outcomes across neighborhoods.

There is substantial variation, with some areas having very low citation rates and others having citation rates close to 100%. This suggests that citation outcomes differ substantially across community areas.

Next, we examine whether this variation can be explained by stop volume. Figure C shows the relationship between citation rate and the number of stops within each community area.

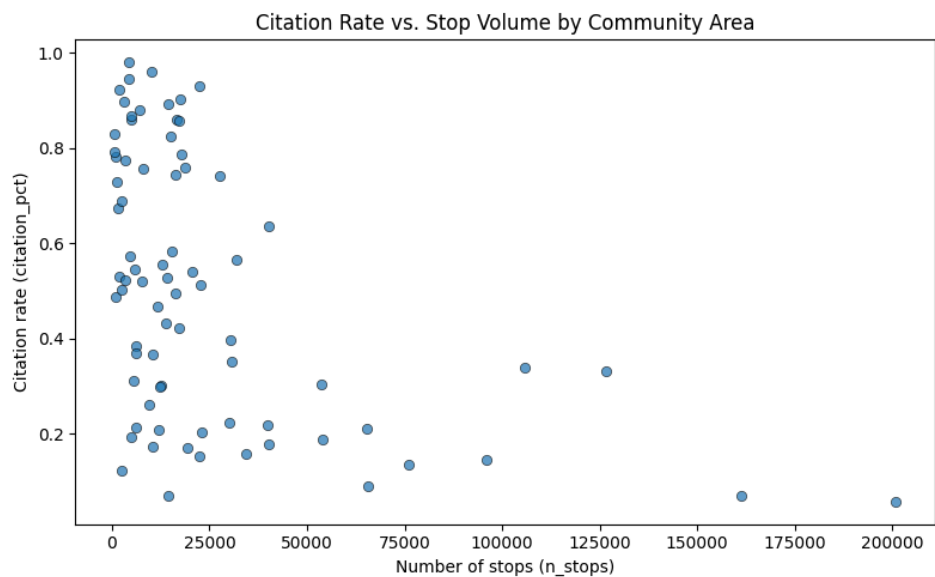


Figure C. Relationship between citation rate and total number of stops by community area. No clear relationship is observed, suggesting that variation in citation rates is not driven solely by stop volume.

There is no clear relationship between stop volume and citation rate, suggesting that differences in citation outcomes are not simply a function of how many stops occur in a community. This suggests that other factors may be contributing to the observed variation.

We then explore how citation rates relate to key socioeconomic variables. Figure D shows the relationship between limited English proficiency and citation rate.

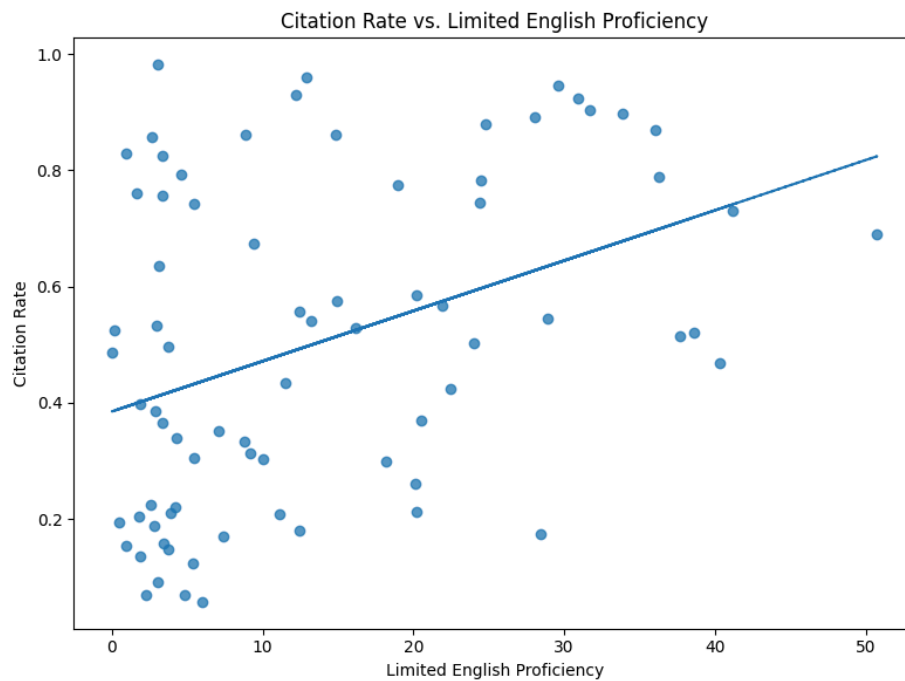


Figure D. Relationship between limited English proficiency and citation rate across community areas. A positive trend suggests that higher levels of limited English proficiency are associated with higher citation rates.

We observe a positive association, suggesting that communities with higher levels of limited English proficiency tend to have higher citation rates.

Figure E shows the relationship between median household income and citation rate.

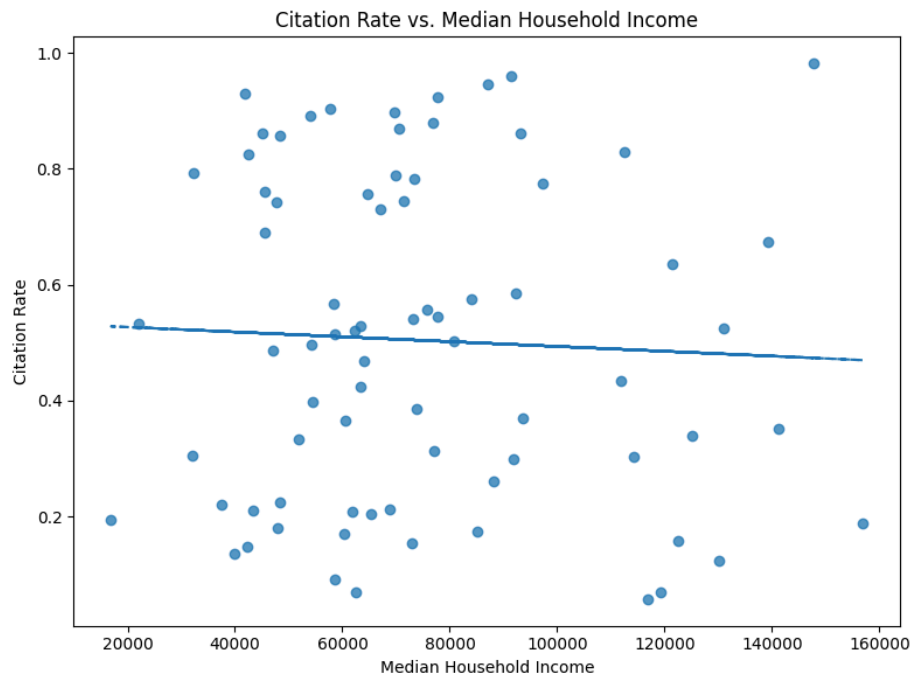


Figure E. Relationship between median household income and citation rate across community areas. A slight negative trend indicates that higher-income communities tend to have lower citation rates.

We observe a slight negative trend, indicating that higher-income communities tend to have lower citation rates. Together, these patterns suggest that socioeconomic characteristics may play a role in explaining variation in citation outcomes.

To further examine relationships among candidate predictors, we computed a correlation matrix (Figure F).

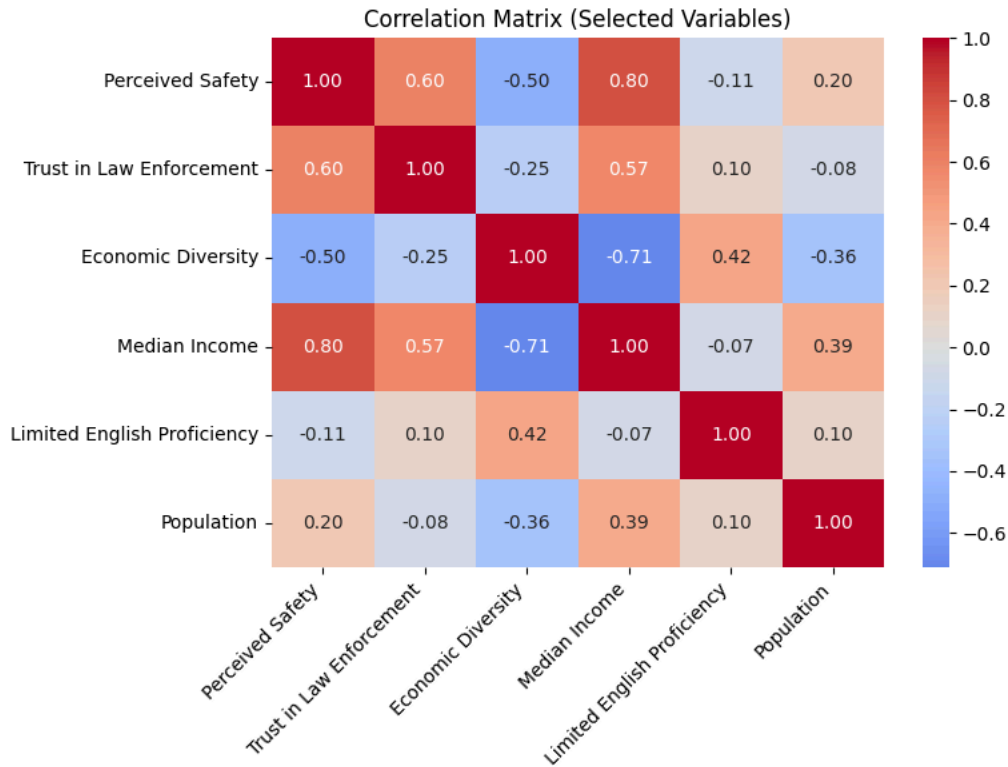


Figure F. Correlation matrix of selected socioeconomic variables used in the regression model. While some moderate correlations are present, extreme multicollinearity is avoided.

As shown in Figure F, the selected variables exhibit moderate correlations but avoid extreme multicollinearity, supporting their inclusion in the regression model.

Overall, the EDA suggests that variation in citation rates is not explained by stop volume alone and may be associated with neighborhood-level socioeconomic factors, motivating the use of regression modeling to further investigate these relationships.

3. Methods and Models

3.1. Technical Approach

To investigate how neighborhood characteristics are associated with traffic stop outcomes, we modeled the probability that a stop results in a citation at the community area level. Because our data were aggregated into counts of citations and non-citations within each community, we used a grouped binomial logistic regression model, which is appropriate for modeling proportions derived

from count data.

Logistic regression was selected because it provides interpretable coefficients, allowing us to quantify both the direction and strength of the relationship between socioeconomic factors and citation likelihood. This aligns with our primary objective of understanding associations rather than maximizing predictive performance.

The model includes several neighborhood-level socioeconomic indicators obtained from the Chicago Health Atlas, including measures of median household income, economic diversity, population size, limited English proficiency, perceived neighborhood safety, and trust in law enforcement. These variables capture multiple dimensions of community context that may influence policing outcomes.

By modeling citation likelihood at the neighborhood level, this approach allows us to identify broader patterns across communities while reducing bias introduced by missing or inconsistent driver-level data.

3.2. Model Training and Tuning

Because the dataset consists of a single aggregated observation for each community area, the model was fit on the full dataset of community-level observations.

Although our primary objective is inference rather than prediction, we explored regularization techniques, including L1, L2, and Elastic Net penalties, to assess the robustness of our results to different modeling specifications. Regularization can shrink coefficients and mitigate multicollinearity, so comparing penalized and unpenalized models allows us to evaluate whether our conclusions depend on these modeling choices.

To support this comparison, we examined weighted mean absolute error (MAE) as a diagnostic measure of model fit. MAE was chosen because it provides a direct and interpretable measure of the average difference between predicted and observed citation rates, weighted by the number of stops in each community. This makes it particularly appropriate in this setting, where communities contribute different amounts of data. Rather than optimizing performance, this metric was used to

confirm that model fit remains consistent across different regularization choices.

Table 1. Comparison of weighted mean absolute error (MAE) across different regularization strategies. Similar performance across most models indicates robustness to penalty choice.

Model Penalty	Weighted MAE
No Penalty	0.152
L1	0.153
L2	0.281
Elastic Net	0.153

Across models, performance was nearly identical to the unpenalized model, indicating that the choice of penalty does not materially affect model fit. The L2 model exhibited substantially worse performance, likely due to instability in estimation, while the remaining models showed nearly identical fit. Given this, we selected the unpenalized logistic regression model as our final model. This choice preserves the interpretability of coefficients, which is central to our objective of understanding how socioeconomic factors are associated with citation likelihood.

3.3. Modeling Assumptions

The final model is a grouped binomial logistic regression applied to community-level aggregated data. Several assumptions underlie this modeling approach.

First, the model assumes that the outcome follows a binomial distribution. In this setting, each observation represents the number of citations out of the total number of stops within a community area, which satisfies the binomial framework.

Second, logistic regression assumes a linear relationship between the predictors and the log-odds of the outcome. While this assumption cannot be fully verified, we used continuous socioeconomic variables and interpreted results as approximate linear relationships at the community level.

Third, the model assumes independence of observations. Because each row corresponds to a distinct community area, observations are largely independent. However, we acknowledge that neighboring communities may exhibit some spatial correlation, which is a limitation of this

analysis.

Fourth, the model assumes that predictors are not highly multicollinear. To assess this, we examined the correlation matrix of candidate variables and removed highly correlated features, such as poverty rate, to reduce redundancy. The remaining variables show moderate correlations but do not indicate severe multicollinearity.

4. Results and Validation

4.1. Results

The final unpenalized logistic regression model achieved a weighted mean absolute error of approximately 0.15, indicating a reasonable level of accuracy in predicting community-level citation rates.

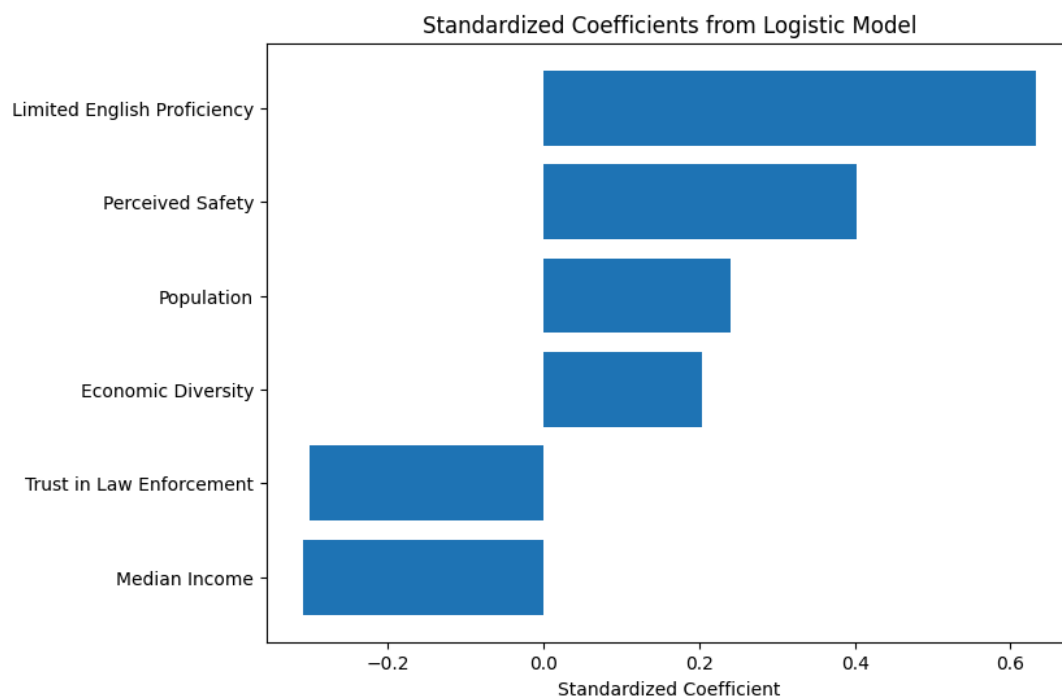


Figure G. Standardized coefficients from the final logistic regression model. Positive values indicate an increase in citation likelihood, while negative values indicate a decrease.

Figure G presents the standardized coefficients from the final model. Standardized coefficients allow for direct comparison of the relative importance of predictors, where positive values indicate

an increase in citation likelihood and negative values indicate a decrease.

Among the variables considered, limited English proficiency exhibits the strongest positive association with citation likelihood, suggesting that communities with higher levels of limited English proficiency tend to have higher citation rates. Perceived neighborhood safety also shows a strong positive association, followed by population and economic diversity, indicating that these factors are associated with higher citation likelihood.

In contrast, median household income and trust in law enforcement show negative associations with citation likelihood, meaning that communities with higher income levels and greater trust in law enforcement tend to have lower citation rates.

All variables included in the model are statistically significant ($p < 0.001$), indicating that these relationships are unlikely to be due to random chance. However, it is important to note that these results reflect associations rather than causal relationships.

Overall, the findings suggest that citation likelihood varies systematically with neighborhood-level socioeconomic conditions and that multiple factors jointly contribute to these differences.

4.2. Validation

To assess the reliability of our findings, we evaluated how well the model's predicted probabilities aligned with observed citation rates using a calibration analysis. Calibration is particularly important in this setting because our goal is to interpret predicted probabilities at the community level rather than simply classify outcomes.

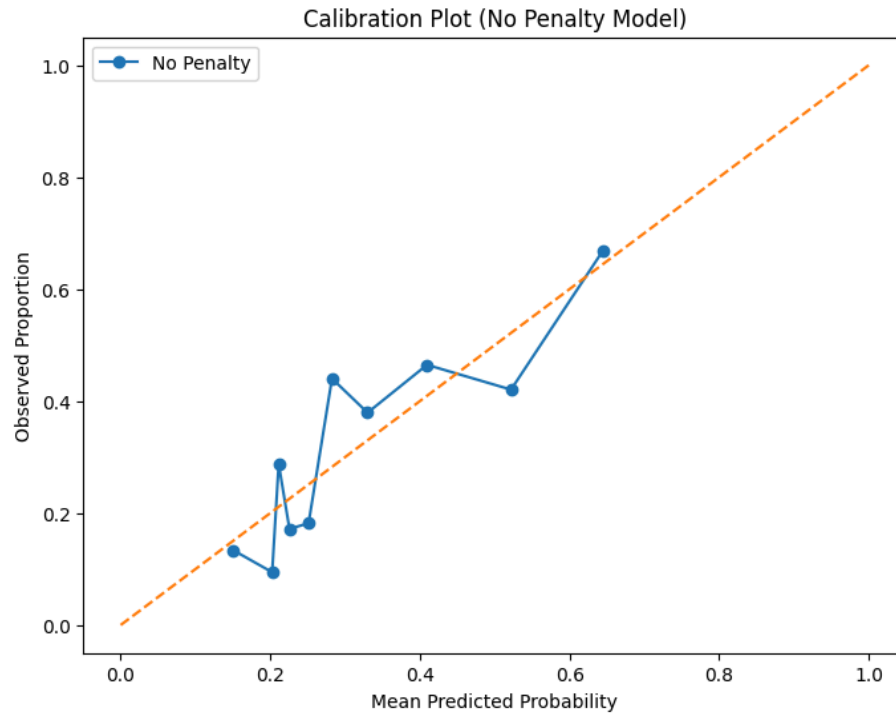


Figure H. Calibration plot for the final logistic regression model. Predicted probabilities closely align with observed outcomes, indicating good calibration across risk levels.

Figure H presents the calibration curve for the final unpenalized logistic regression model. The model closely follows the 45-degree reference line, indicating that predicted citation probabilities are well-aligned with observed proportions. This suggests that the model provides well-calibrated probability estimates across different levels of predicted risk.

In addition to calibration, the relatively low weighted mean absolute error (approximately 0.15) indicates that predicted citation rates are, on average, close to the observed rates across communities. Together, these results demonstrate that the model produces stable and reliable estimates.

Overall, these validation checks support the use of the model for interpreting relationships between neighborhood-level socioeconomic factors and citation likelihood.

5. Discussion and Conclusion

5.1. Discussion of Impact

This analysis provides insight into how neighborhood-level socioeconomic factors are associated

with citation likelihood in traffic stops. Rather than being randomly distributed, citation outcomes appear to vary systematically across communities, suggesting that broader social and economic conditions may play a role in policing patterns. These findings highlight the importance of considering contextual, neighborhood-level factors when evaluating policing outcomes.

In addition to the substantive findings, this project highlights the importance of careful data and modeling choices when working with real-world data. Aggregating observations to the community level allowed us to focus on broader patterns rather than individual stops, and choosing an interpretable model made it possible to clearly understand how each factor relates to citation likelihood. These decisions were important for ensuring that the results are both meaningful and easy to interpret, making them more accessible to a lay audience.

A key takeaway from this analysis is that no single factor explains variation in citation likelihood. Instead, multiple socioeconomic characteristics jointly contribute to these differences across communities. This reinforces the importance of taking a holistic approach when analyzing complex social outcomes.

One important methodological decision was to aggregate the data to the community level due to limitations and inconsistencies in driver-level information. By shifting the analysis to a neighborhood-level perspective, we were able to reduce noise in the data and produce more stable and interpretable results. This approach serves as an example of how adapting the structure of the data can lead to more meaningful insights in applied data science problems.

5.2. Limitations and Future Work

This analysis has several limitations. First, the data were aggregated at the community level, which improves stability but limits insight into individual-level variation in traffic stop outcomes. Second, the results are based on observational data, so the relationships identified are associative rather than causal.

The analysis is also limited by data availability. Important factors such as officer behavior and detailed stop context were not included, which may influence citation outcomes. Additionally, the socioeconomic data lack consistent temporal resolution, restricting the ability to examine changes

over time.

Finally, the model assumes independence across community areas, though nearby communities may share similar characteristics, introducing potential spatial correlation.

Future work could address these limitations by incorporating more detailed, time-varying data and applying spatial modeling techniques to better capture relationships across communities, potentially identifying patterns relevant to policymaking.

6. Reproducibility

6.1. Data Access

The data used in this project are publicly available. Traffic stop data were obtained from the [Stanford Open Policing Project](#), which provides detailed records of police stops in Chicago. Socioeconomic and neighborhood-level variables were sourced from the [Chicago Health Atlas](#).

To reproduce this analysis, users can download the raw datasets directly from these sources. The traffic stop dataset can be accessed through the Stanford Open Policing Project website, while the neighborhood-level data can be obtained from the Chicago Health Atlas website by selecting the relevant indicators and exporting the data.

After downloading the datasets, users can reproduce the data cleaning and merging process by running the preprocessing notebook provided in the project's GitHub repository.

6.2. GitHub Repository

The full codebase is available at: <https://github.com/utexas-SDS-357/sds357-project-sp26-analysts>

To reproduce the analysis:

1. Download the raw datasets as described in Section 6.1 (Data Access).
2. Clone or download the GitHub repository to your local machine.
3. Navigate to the `data_wrangling` folder and run the notebooks in the following order:
 - `data_wrangling.ipynb` to clean and merge datasets spatially by community area

- `community_area_grouping.ipynb` to produce the final aggregated dataset
- 4. Navigate to the Modeling folder and run:
 - `grouped_binomial_logistic.ipynb` to reproduce the final model results and coefficient interpretations

All steps required for data preprocessing, modeling, and visualization are included in the repository to ensure full reproducibility of the analysis.

7. References

- **American Civil Liberties Union Report:**
<https://www.aclu-il.org/publications/2015-stop-and-frisk-report/>
- **Academic Paper (Duke University Press Demography):** Kirk D. S. (2008). The neighborhood context of racial and ethnic disparities in arrest. *Demography*, 45(1), 55–77.
<https://doi.org/10.1353/dem.2008.0011>
- **Academic Paper (PNAS):** Xu, W., Smart, M., Tilahun, N., Askari, S., Dennis, Z., Li, H., & Levinson, D. (2024). The racial composition of road users, traffic citations, and police stops. *Proceedings of the National Academy of Sciences of the United States of America*, 121(24), e2402547121. <https://doi.org/10.1073/pnas.2402547121>